

Suction Lines 521

Operation Manual

Read this manual completely and keep it!

Imprint

**Suction Lines 521
Operation Manual**

Version 1.0

Publisher

ALLDOS Eichler GmbH
Reetzstraße 85 • 76327 Pfinztal (Söllingen)
Postfach 1160 • 76317 Pfinztal
Germany

Tel. +49 72 40 61-0 / Fax. +49 72 40 61-211

Mail: alldos.de@alldos.com
Internet: www.grundfosalldos.com

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1 General

1.1 General Information

This technical information contains all the information required for starting up and handling the described suction lines.

If you require further information or if any problems arise that are not discussed in detail in this document, contact Grundfos Alldos directly for more information.

1.2 Using this document

The safety instructions **WARNING**, **CAUTION** and **NOTE** have the following meanings:



Warning

Risk of accidents and injury!



Attention

Risk of malfunction or damage to the device!



Note

There is an exceptional feature.

1.3 Guarantee

A guarantee claim in the sense of our general conditions of sale and delivery can only be recognised if:

- The component has been used according to the information in this technical information,
- The component is not subjected to improper use,
- Installation, service, and repairs are only carried out by authorised and qualified personnel,
- Only original spare parts are used for repairs

The safety instructions specified in this document must be observed at all times.

2 Safety

The component has left the factory in a safe error-free state.

In order to maintain this state and ensure safe operation, the user must observe the instructions and warning notes provided in this technical information.

If safe operation is no longer possible, the component must be shut down and secured against unintentional operation. This is the case:

- If the component is visibly damaged
- If the component no longer seems operational
- Following long periods of storage under unfavourable conditions

2.1 Using the Component

The suction line systems described here are used in connection with Grundfos Alldos dosing pumps.

Possible applications include:

- Installation in Grundfos Alldos dosing tanks
- Installation in transport tanks or canisters
- Installation in plant-specific dosing tanks

2.2 Obligations of the owner

The owner of the plant is responsible for

- Training of operating personnel
- Arranging regular maintenance.



Warning

Other applications are considered as non-approved, and are not permissible. Grundfos Alldos accepts no liability for any damage resulting from incorrect use.

2.3 Avoidance of danger



Warning

When dosing dangerous media, observe the corresponding safety precautions!



Warning

***When working with connections and suction lines:
Always wear protective clothing (safety glasses and gloves)!***



Attention

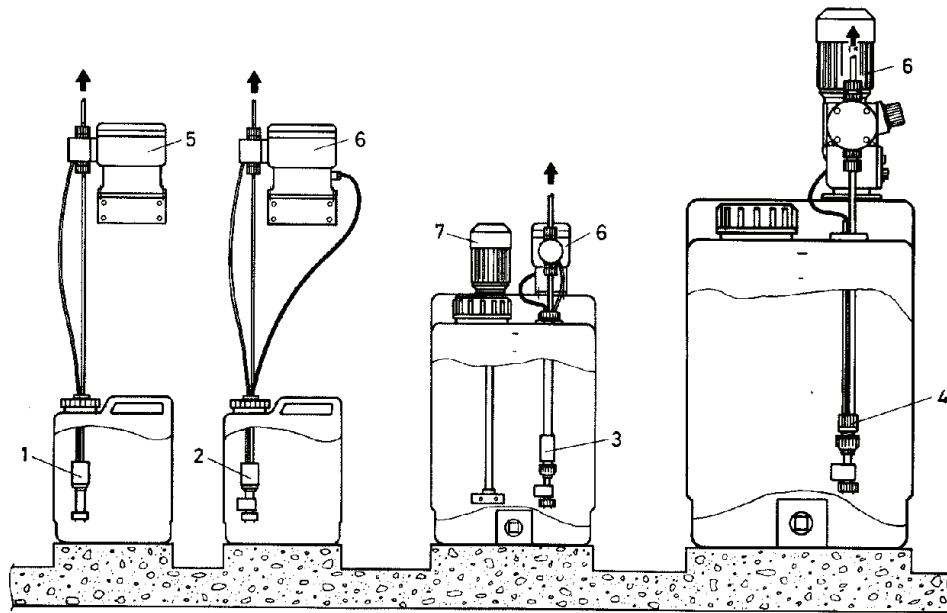
Please observe the chemical resistance of dosing media with regard to materials.

3 Construction and Function

Components of Grundfos Alldos suction lines are:

- line
 - rigid (pipe) or
 - flexible (tube)
- Foot valve (ball check valve) with sieving filter
 - spring-loaded (tantalum spring) for DN 8
- Reed contact float switch for suction lines designed for pumps with empty signal electronics.
 - integrated in the valve body
 - float with encapsulated magnet is installed around the valve body, and rises and falls with liquid level

According to the application, different materials can be chosen.

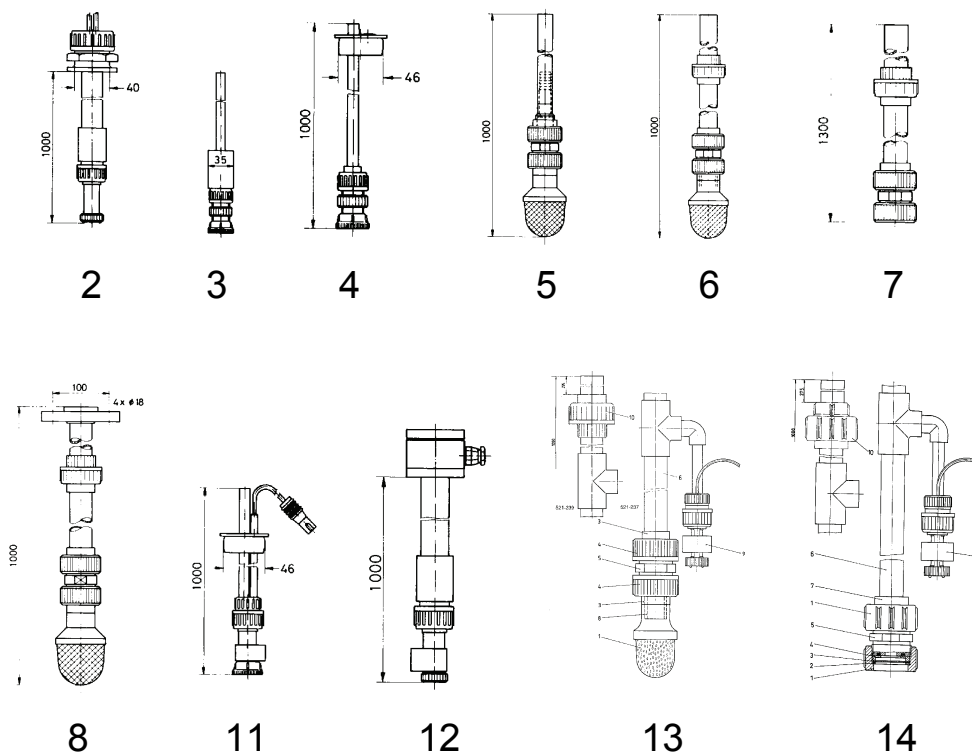


- 1 Flexible suction line
- 2 Flexible suction line with reed switch
- 3 Rigid suction line, with reed switch for dosing tank with agitator
- 4 Rigid suction line, with reed switch
- 5 Standard dosing pump
- 6 Dosing pumps with empty signal electronics
- 7 Agitator

4 Technical Data

Order No.	DN	Line type	Length*	Materials			Supplementary equipment	Fig.
				Valve body	Valve seat	Valve ball		
Suction lines without reed contact float switch								
521-015	8	PVC hose 6/12	1 m	PVC	Viton	Duran glass	Loading stone of porcelain	3
521-016	8	PVC pipe 12 x 1.4	1 m	PVC	Viton	Duran glass	PE plug-on-cap for tank installation	4
521-027	8	PVDF pipe 12 x 1	1 m	PVDF	PTFE	PTFE	PE plug-on-cap for tank installation	4
521-041	10	PP pipe 16 x 2	1 m	PPH	Viton	Duran glass	PE plug-on-cap for tank installation	4
521-033	20	PVC hose 13/20	1 m	PVC	PTFE, 20% GF	Duran glass		5
521-034	20	PVC hose 19/27	1 m	PVC	PTFE, 20% GF	Duran glass		5
521-036	20	PVC pipe 25 x 2.7	1 m	PVC	PTFE, 20% GF	Duran glass		6
521-040	20	PP pipe 25 x 2.3	1 m	PP	PTFE, 20% GF	Duran glass		7
521-028	20	PVDF pipe 25 x 1.9	1 m	PVDF	PTFE, 20% GF	PTFE		7
521-037	32	PVC pipe 40 x 3	1 m	PVC	PE	Soda glass	Connection flange DN 32 according to DIN	8
Suction lines with reed contact float switch for dosing pumps with empty signal electronics, cpl. with 1.5 m cable and plug for the socket of the dosing pump								
521-017	8	PVC pipe 12x1.4	1 m	PVC	Viton	Duran glass	PE plug-on-cap for tank installation	11
521-020	8	PVDF pipe 12x1	1 m	PVDF	PTFE	PTFE	PE plug-on-cap for tank installation	11
521-018	8	PVC pipe 12x1.4	0,7 m	PVC	Viton	Duran glass	height adjustable, PE plug-on-cap for tank installation	11
521-416	8	like 521-016, cable length of float switch 10 m						
521-237	20	PVC pipe 25 x 1.9	1 m	PVC	PTFE, 25% GF	Duran glass	height adjustable, without screwing	13
521-239	20	PVC pipe 25 x 1.9	1 m	PVC	PTFE, 25% GF	Duran glass	height adjustable, with screwing	13
521-238	20	PVDF pipe 25 x 1.9	1 m	PVDF	PTFE, 25% GF	PTFE	height adjustable	14
521-240	20	PVDF pipe 25 x 1.9	1 m	PVDF	PTFE, 25% GF	PTFE	height adjustable, with plug and screwing	14
521-241	20	PP pipe 25 x 2.3	1 m	PP	PTFE, 25% GF	Duran glass	height adjustable, without plug and screwing	14
338-204	empty signal probe with reed contact float switch, material PVC, cpl. with terminal box for fastening on the dosing tank, for use with suction lines DN 20 and DN 32 for empty indication							12

* The lengths of hoses or tubes quoted in the table or illustrations are standard lengths. If ordered together with Grundfos Alldos dosing tanks the length of the suction line is adapted to the tank.



Measurements in mm.

5 Installation and Operation

- The suction lines must be installed in the dosing tank with the foot valve remaining a few centimetres above the tank or container bottom.



Warning

Wear protective gloves in case of aggressive media!

- Connect the suction line with the dosing pump.



Attention

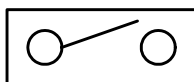
Observe the Technical Information of the dosing pump.



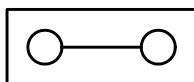
Warning

Observe the safety precautions for handling the chemicals!

- Possible applications for suction lines with float switch:
- Function "close contact empty signal":



- the float must be fitted at the valve body with the symbol on top of the float.
- Function "break contact empty signal":



- the float must be fitted at the valve body with the symbol on top of the float.



Note

The Grundfos Alldos dosing pumps with empty signal electronics are principally adjusted for the function "close contact empty signal". In order to change the poles, the float can be removed, turned around and fitted back again.

6 Maintenance

If the suction capacity decreases:

- Remove foot valve and clean it.
- replace wearing parts (valve ball, valve spring, valve seat, O-rings) if necessary.

6.1 Spare Parts

Suction line	Order No.			
	Valve spring	Valve ball	Valve seat	O-rings
521-015	10.141-4	10.139	10.269-42	52.105-2 (2 pieces)
521-016	10.141-4	10.139	10.269-42	52.105-2 (2)
521-027	10.141-4	10.073	10.269-41	52.105-2 (3)
521-041	10.141-4	10.139	10.269-42	52.105-2 (3)
521-033	-	10.227	10.6505-400	52.146 (1), 52.141 (2)
521-034	-	10.227	10.6505-400	52.146 (1), 52.141 (2)
521-036	-	10.227	10.6505-400	52.146 (1), 52.141 (2)
521-040	-	10.227	10.6505-400	52.141 (2), 52.146 (1)
521-028	-	10.1589	10.6505-405	52.146 (1), 52.141 (2)
521-037	-	10.468	10.2153-41	52.230 (2)
521-017	10.141-4	10.139	10.269-42	52.298 (1)
521-020	10.141-4	10.073	10.269-41	52.298 (1)
521-018	10.141-4	10.139	10.269-42	52.298 (1)
521-416	10.141-4	10.139	10.269-42	-
521-237	-	10.227	10.6505-400	52.146 (1), 52.141 (2), 52.127 (2), 52.298 (1)
521-239	-	10.227	10.6505-400	52.146 (1), 52.141 (2), 52.127 (2), 52.298 (1), 52.133 (1)
521-238	-	10.1589	10.6505-405	52.146 (1), 52.141 (2), 52.127 (2), 52.298 (1)
521-240	-	10.1589	10.6505-405	52.146 (1), 52.141 (2), 52.127 (2), 52.298 (1)
521-241	-	10.227	10.6505-400	52.146 (1), 52.141 (2), 52.127 (2), 52.298 (1)

